



ALLIANCE
UNIVERSITY
Private University established in Karnataka State by Act No. 84 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi

NAAC
GRADE A+
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Data Cleaning and Preparation using Power Query

Submitted By:

Name: ATHUL K KOSHY

Reg. No.: 2411021240048

Date: 10-09-2026

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 4

Data Cleaning and Preparation using Power Query

[GITHUBLINK](https://github.com/athul-hue/BUSINESS-ANALYTICS-PROJECT) <https://github.com/athul-hue/BUSINESS-ANALYTICS-PROJECT>

Problem Statement

A café has collected customer transaction data for sales analysis. However, the dataset contains several data-quality problems, including missing values, duplicate transactions, incorrect entries, inconsistent text, and invalid values such as “ERROR” and “UNKNOWN.” These issues can lead to incorrect calculations and unreliable reports. Therefore, Power Query is used to clean, transform, standardize, and prepare the dataset for accurate analysis in Power BI.

Methodology / Steps Followed

Step 1: Data Collection

- Dataset sourced from Kaggle: “Cafe Sales – Dirty Data for Cleaning Training” (dirty_cafe_sales.csv)
- The raw file contains 8 columns – Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date – across 1,000+ rows
- The dataset is intentionally “dirty,” containing missing values, invalid placeholder text (“ERROR”, “UNKNOWN”), and inconsistent entries to simulate real-world raw data.

- Step 2: Data Import
- Opened Power BI Desktop.
- Selected Get Data → Text/CSV.
- Imported the dirty_cafe_sales.csv file.
- Verified the delimiter, file encoding, column names, and number of records.
- Selected Transform Data to open the dataset in Power Query Editor

Data Quality Checking

- Opened the View tab in Power Query.
- Enabled:
 - Column Quality
 - Column Distribution
 - Column Profile
- Used these features to identify errors, empty values, and unusual entries.
- Found invalid values such as “ERROR” and “UNKNOWN.”
- Identified missing values in fields such as Item, Payment Method, Location, Quantity, Price Per Unit, and Transaction Date.

Step 4: Cleaning Invalid and Missing Values

- Replaced “ERROR” and “UNKNOWN” with null.
- Checked blank and null values in important columns.
- Filled suitable missing text values with “Not Specified” where required.
- Removed records where missing information made the transaction unsuitable for analysis.
- Checked the dataset again to ensure that invalid values were properly handled.

Step 5: Correcting Data Types

- Changed Transaction ID to Text.
- Changed Item to Text.
- Changed Payment Method to Text.
- Changed Location to Text.
- Changed Quantity to Whole Number.
- Changed Price Per Unit to Decimal Number.
- Changed Total Spent to Decimal Number.
- Changed Transaction Date to Date.

Step 6: Creating the Correct Total Spent

- Checked the existing Total Spent values for errors and missing entries.
- Created a new calculated column using:

Total Spent = Quantity × Price Per Unit

- Used the calculated value to replace incorrect or missing Total Spent values.
- Removed the old incorrect Total Spent column.
- Renamed the new column to Total Spent.

Step 7: Removing Duplicate Records

- Checked the dataset for duplicate transactions.
- Used Transaction ID as the key field for identifying duplicate records.
- Removed duplicate Transaction IDs to ensure that each transaction appeared only once.

Step 8: Standardizing Text and Dates

- Used Trim to remove unnecessary spaces from text fields.
- Used Clean to remove unwanted characters.
- Standardized text values to maintain consistency.
- Converted all transaction dates into a consistent Date format.

Step 9: Final Validation

- Rechecked Column Quality and Column Profile after completing the cleaning process.
- Confirmed that invalid values and duplicate records had been reduced or removed.
- Verified that numerical columns contained appropriate numeric values.
- Checked that the calculated Total Spent values were correct.
- Ensured that the cleaned dataset was suitable for reporting.

Step 10: Loading the Clean Dataset

- Selected Close & Apply in Power Query.
- Loaded the cleaned data into the Power BI data model.
- The final dataset was now ready for creating sales reports, dashboards, charts, and further analysis.

▼ Select a data source or start with a blank report

Blank report

OneLake catalog

Excel workbook

SQL Server

Learn with sample data

Get data from other sources

▼ Recommended

Getting started

Intro—What is Power BI? [?](#)

Recent Shared with me

Name	Location
Ba-Exp4.pbix	C:\Users\athul\Downloads\POWER BI\Ba-Exp4.pbix

Output

dirty_cafe_sales.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8
Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16
TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023-07-19
TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
TXN_2602893	Smoothie	5	4.0	20.0	Credit Card		2023-03-31
TXN_4433211	UNKNOWN	3	3.0	9.0	ERROR	Takeaway	2023-10-06
TXN_6699534	Sandwich	4	4.0	16.0	Cash	UNKNOWN	2023-10-28
TXN_4717867		5	3.0	15.0		Takeaway	2023-07-28
TXN_2054365	Sandwich	5	4.0	20.0		In-store	2023-12-31
TXN_2548360	Salad	5	5.0	25.0	Cash	Takeaway	2023-11-07
TXN_3051279	Sandwich	2	4.0	8.0	Credit Card	Takeaway	ERROR
TXN_7619095	Sandwich	2	4.0	8.0	Cash	In-store	2023-05-03
TXN_9437049	Cookie	5	1.0	5.0		Takeaway	2023-06-01
TXN_8915701	ERROR	2	1.5	3.0		In-store	2023-03-21
TXN_2847255	Salad	3	5.0	15.0	Credit Card	In-store	2023-11-15
TXN_3765707	Sandwich	1	4.0	4.0			2023-06-10
TXN_6769710	Juice	2	3.0	6.0	Cash	In-store	2023-02-24
TXN_8876618	Cake	5	3.0	15.0	Cash	ERROR	2023-03-25

The data in the preview has been truncated due to size limits.

Extract Table Using Examples Load Transform Data Cancel

Transaction ID	Item	Quantity	Payment Method	Location	Transaction Date	Price Per Unit	Total Spent
TXN_1961373	Coffee	2	Credit Card	Takeaway	08-09-2023	2	4
TXN_4977031	Cake	4	Cash	In-store	16-05-2023	3	12
TXN_4271903	Cookie	4	Credit Card	In-store	19-07-2023	1	4
TXN_7034554	Salad	2		null	27-04-2023	5	10
TXN_3160411	Coffee	2	Digital Wallet	In-store	11-06-2023	2	4
TXN_2602893	Smoothie	5	Credit Card	null	31-03-2023	4	20
TXN_4433211		3		null	06-10-2023	3	9
TXN_6699534	Sandwich	4	Cash		28-10-2023	4	16
TXN_4717867		5		null	28-07-2023	3	15
TXN_2064365	Sandwich	5		null	31-12-2023	4	20
TXN_2548360	Salad	5	Cash	Takeaway	07-11-2023	5	25
TXN_7619095	Sandwich	2	Cash	In-store	03-05-2023	4	8
TXN_9437049	Cookie	5		null	01-06-2023	1	5
TXN_8915701		2		null	21-03-2023	1.5	3
TXN_2847255	Salad	3	Credit Card	In-store	15-11-2023	5	15
TXN_3765707	Sandwich	1		null	10-06-2023	4	4
TXN_6769710	Juice	2	Cash	In-store	24-02-2023	3	6
TXN_8876618	Cake	5	Cash		25-03-2023	3	15
TXN_3709394	Juice	4	Cash	Takeaway	15-01-2023	3	12
TXN_3567645	Smoothie	4	Credit Card	Takeaway	30-03-2023	4	16
TXN_5132361	Sandwich	3	Digital Wallet	Takeaway	01-12-2023	4	12
TXN_2616390	Sandwich	2		null	18-09-2023	4	8
TXN_9400181	Sandwich	5	Cash	In-store	03-06-2023	4	20
TXN_7958992	Smoothie	3		null	13-12-2023	4	12
TXN_5183041	Cookie	5	Credit Card	In-store	20-04-2023	1	

- **Power Query Editor:** Displays the imported `dirty_cafe_sales` dataset with its eight main fields: Transaction ID, Item, Quantity, Payment Method, Location, Transaction Date, Price Per Unit, and Total Spent. The data-profiling features are enabled to inspect the quality of the dataset.
- **Column Quality Analysis:** Shows the percentage of valid, error, and empty values for each column. This helped identify columns containing a high number of missing or incorrect records, particularly Payment Method and Location.
- **Raw Data Preview:** Shows the original café sales data before transformation. The preview contains incomplete records, blank cells, and invalid entries such as “ERROR” and “UNKNOWN.”
- **Applied Steps:** Shows the sequence of Power Query transformations performed during the cleaning process. The steps include removing the incorrect Total Spent column, creating a corrected calculation, and renaming the new column to Total Spent

Conclusion

The café sales dataset was successfully cleaned and transformed using Power Query. The cleaning process helped identify and resolve missing values, invalid entries, duplicate transactions, incorrect data types, and inconsistent formats. The Total Spent field was also corrected using Quantity and Price Per Unit. As a result, the raw dataset was converted into a more reliable and structured dataset that can be used for Power BI reports and sales analysis.

Learning Outcomes

- Learned how to import and prepare CSV data using Power Query.
- Learned how to identify data-quality problems using Column Quality, Column Distribution, and Column Profile.
- Gained experience in handling missing and invalid values.
- Learned how to replace unwanted values such as “ERROR” and “UNKNOWN.”
- Understood how to assign the correct data types to different columns.
- Learned how to create calculated columns using existing fields.
- Practiced removing duplicate records from transaction data.
- Learned how to clean text fields and standardize date formats.
- Understood how Applied Steps make the cleaning process repeatable and easy to track.

Business Insights

- Missing information in Payment Method and Location can reduce the accuracy of sales reports and make customer behavior analysis difficult.
- The presence of “ERROR” and “UNKNOWN” values indicates that the café should improve its data-entry and transaction-recording process.
- Incorrect Total Spent values can affect revenue calculations and lead to misleading business decisions.
- Duplicate transactions can result in sales being counted more than once, which may incorrectly increase reported revenue.
- Standardizing transaction data improves the reliability of future dashboards and reports.
- A structured Power Query workflow reduces manual data-cleaning work and makes the process easier to repeat whenever new café transaction data is received.